

Detecting Ransomware Before Encryption: AI-Based Network Anomaly Detection

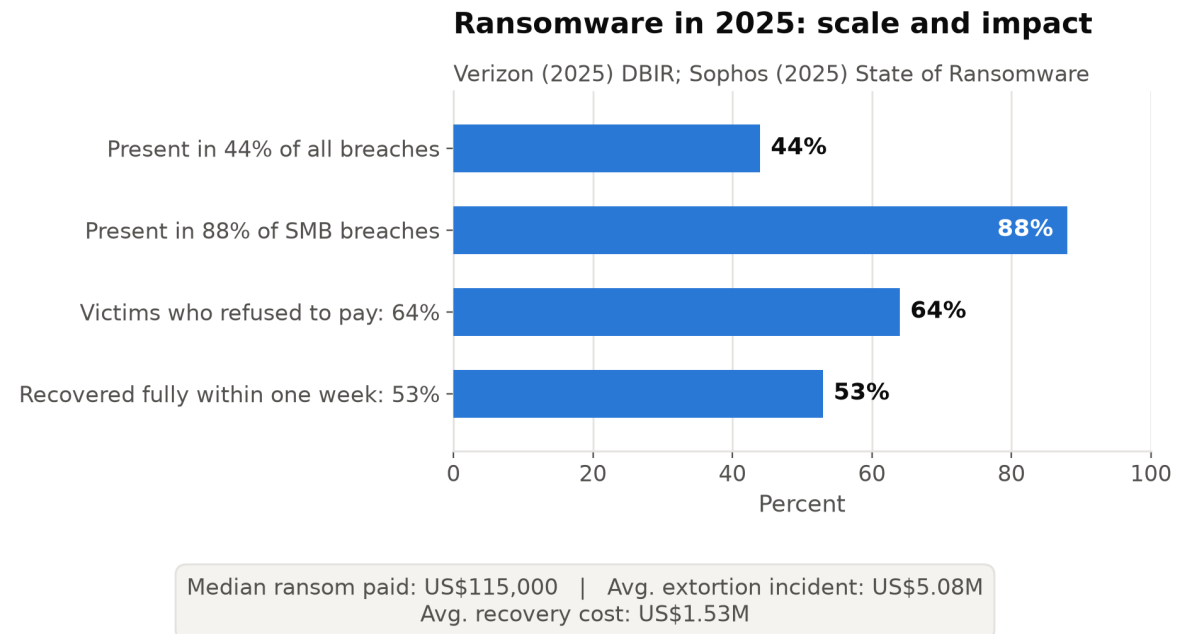
Balancing Automated Detection with AI Governance

Introduction: Why Ransomware Needs Earlier Detection

- Ransomware appeared in 44% of data breaches in the 2025 DBIR, up 37% year over year (Verizon, 2025).
- The median ransom payment was US\$115,000, while 64% of victims refused to pay (Verizon, 2025).
- When attackers disclosed the incident, the average extortion or ransomware breach cost reached \$5.08 million (IBM, 2025).
- Question: can AI flag suspicious network behavior before encryption starts?

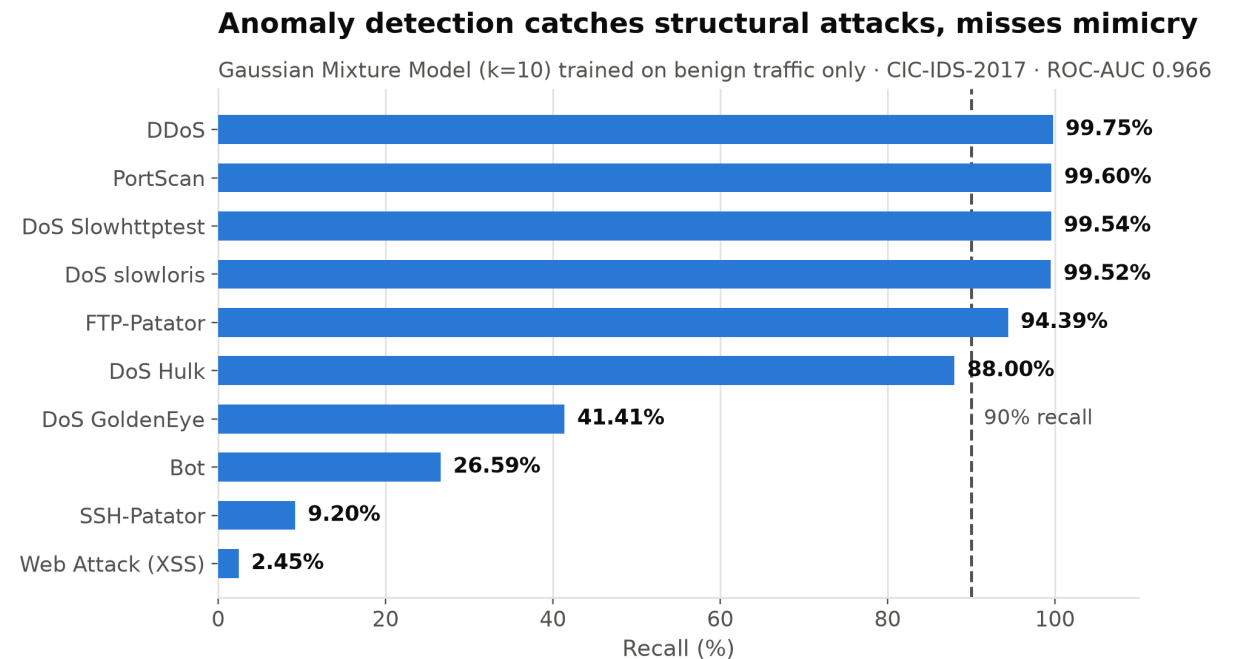
Threat Overview: Impact and Pre-Encryption Signals

- Ransomware appeared in 44% of breaches overall and 88% of SMB breaches; attacks rose 37% year over year (Verizon, 2025).
- The median payment was US\$115,000. An extortion or ransomware breach disclosed by the attacker averaged \$5.08 million (Verizon, 2025; IBM, 2025).
- Ransomware deployment is often preceded by command-and-control activity and lateral movement, creating a window for network detection (CISA et al., 2023).



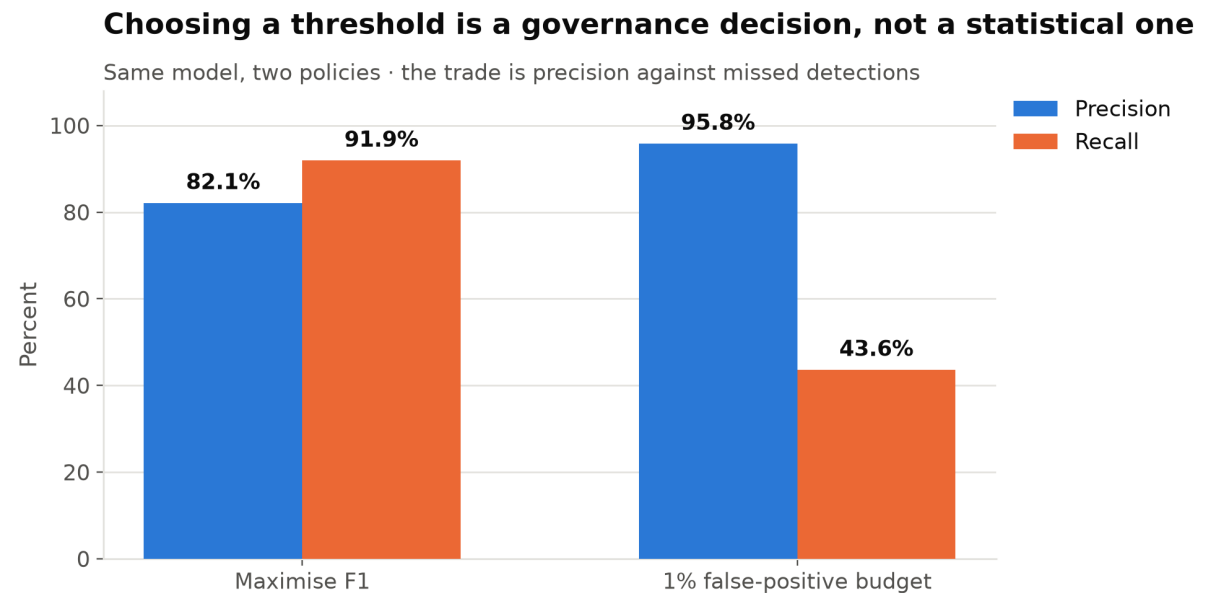
AI Solution: Network Anomaly Detection

- Chapter 5 models normal network traffic with Gaussian density and flags low-probability flows as anomalies (Parisi, 2019).
- A benign-only Gaussian Mixture Model ($k = 10$) reached ROC-AUC 0.9655 on 627,314 held-out flows.
- Recall reached 99.75% for DDoS and 99.60% for PortScan, but only 26.59% for Bot and 9.20% for SSH-Patator.
- The evidence supports an early-warning layer, not a stand-alone ransomware control.



Current Trend: AI Governance for Automated Response

- Threshold choice determines how many threats the system misses and how many benign flows it blocks.
- At a 1% false-positive budget, my GMM delivered 95.8% precision but only 43.6% recall.
- 63% of breached organizations lacked an AI governance policy or were still developing one; one in five reported a shadow-AI breach (IBM, 2025).
- NIST AI RMF provides a structure for documenting risk, monitoring performance, and assigning human oversight (NIST, 2023).



Conclusion: Early Warning with Human Oversight

- Anomaly detection can expose network behavior that precedes ransomware encryption, but coverage varies sharply by attack family.
- My lab showed that model choice and threshold selection are security decisions, not just technical tuning.
- In practice, I would document the threshold, retain an audit trail, and require a named analyst to authorize high-impact actions.

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